

MATHEMATICAL METHODS IN BIOENGINEERING
TEST # 5

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The following data relate the average daily exposure time to UV-B radiation (measured in hours over a biological control period) to the proportion of laboratory mice that develop benign skin lesions.

Daily UV-B Exposure (Hours)	Proportion with Skin Lesions
1.0	0.0210
2.0	0.0450
3.0	0.0880
4.0	0.1520
5.0	0.2340
6.0	0.3550
7.0	0.5120
8.0	0.6890
9.0	0.8210
10.0	0.9450

- P1. [13 p]** Adapt the Python code you have on your computer to find the values for α and β of the least squares line of best fit $Y = \alpha + \beta x$ for the given dataset, where x represents the daily UV-B exposure hours and Y represents the proportion of mice developing skin lesions.
- P2. [13 p]** Create a single figure containing both a scatter plot of the original dataset points and a plot of your linear regression model $Y = \alpha + \beta x$.
- P3. [13 p]** Estimate the probability that a mouse exposed to an average of 6.4 hours of daily UV-B radiation will develop skin lesions.
- P4. [6 p]** To the figure of problem P2., add a point, ideally with a different colour, showing your projection of problem P3.
- P5. [15 p]** Suppose you propose a different model: a linear model with the same intercept α as the one in P1–P4, but in which the probability that a mouse develops skin lesions grows 70% faster with respect to x compared to the original model. Plot this new model in the same figure with the scatter plot of the dataset. Using this new model, estimate the number of hours of daily UV-B exposure required for the probability of developing a skin lesion to reach 97%.